CS 4331-001

# Report for GenAI Project: Comparing 3D-GAN (Traditional Gen vs ConvNeXt)

at Texas Tech University


Samuel Kalu, C.E. R#11819437

Texas Tech University

Computer Science Department


Dec 2025

Notebook with visualizations: https://github.com/SamKa1u/GenAI-3d-GAN/blob/main/GenAI_Project.ipynb

## Abstract

Three-dimensional object generation remains a challenging task in generative modeling, with significant implications for virtual reality content creation, 3D printing applications, and automated asset generation. At the same time, many traditional 3D-GAN architectures only rely on CNN backbones with BatchNorm, leaving the door open for applying modern architectural innovations to enhance volumetric synthesis quality. This project investigates the effectiveness of adapting ConvNeXt architectural principles to 3D generative adversarial networks for voxel-based synthesis. Specifically examining whether replacing BatchNorm with LayerNorm in ConvNeXt-style generator blocks will improve generation quality and diversity.

## Acknowledgements

This project would not have been possible without the contribution, support, or resources of:

- Texas Tech University: CS Department
- IEEE
- Xchhuang [1]
- Atakanerdogan305 [2]

It is affirmed that the work discussed in this report was done for this class and has not been, nor will be submitted for credit in another class.

**Table of Contents**

**List of Figures**



**List of Tables**

## 1. Introduction

By learning to map low-dimensional random noise to high-dimensional output spaces, GANs have shown remarkable capabilities in generative modeling. This enables the synthesis of complex, realistic data samples. Originally developed for high resolution image generation, GAN architectures have been repurposed for 3D reconstruction, allowing for the creation of new ways to gather volumetric data. 3D generative models offer significant practical value in creating digital assets for Virtual Reality (VR) experiences, where traditional 3D modeling software presents substantial complexity barriers and skilled professionals remain scarce and costly. Automated 3D object generation has the chance to democratize asset creation across the virtual reality market space. The Prevailing paradigm for 3D-GAN implementations leverages Convolutional Neural Networks (CNNs), however the success of Vision Transformers in discriminative tasks has prompted a deeper look into attention–based mechanisms for generative modeling. ConvNeXt, a modernized CNN architecture incorporating design principles from Vision Transformers, has demonstrated competitive or superior performance compared to ViTs on standard image classification benchmarks. This study investigates the effectiveness of adapting ConvNeXt architectural principles to 3D-GAN frameworks, hypothesizing that the modernized design may yield improvements in volumetric object synthesis.

## 2. Experimental Results and Analysis

The performance of two generator architectures was evaluated using two widely adopted metrics for generative model assessment. Namely, the Fréchet Inception Distance (FID) and the Inception Score (IS). These metrics leverage features extracted from a pretrained Inception network to quantify the diversity and determine the quality of generated samples. The dataset for generated images totaled 500 samples for each generator while the real dataset held 10,668 samples. Below, Table 1 presents the comparative performance of the original DCGAN architecture against the presented ConvNeXt variant.

| Model | FID (↓) | IS Mean (↑) | IS Std |
|---|---|---|---|
| DCGAN (Baseline) | **419.49** | 1.319 | 0.038 |
| ConvNeXt-DCGAN | 423.73 | **1.336** | 0.043 |

Table 1: Quantitative Comparison of Generator Architectures.

The original DCGAN architecture received a marginally lower FID score (419.49) vs the ConvNeXt variant (423.73). This represents a difference of 1.01% relative increase. The FID score measures the distance between feature distributions of fake and real samples. Lower values show greater similarity to the real data distribution. The increase in FID score suggests the architectural modifications were not effective in increasing distributional alignment with the real data distribution. Conversely, the ConvNeXt variant showed a higher inception score (1.336 ± 0.043) compared to the original's (1.319 ± 0.038). This score evaluates both the quality and diversity of generated samples, by measuring the KL divergence between conditional and marginal class distributions. The 1.29% improvement suggests ever so slightly enhanced sample diversity and quality; however, the difference is still within the margins of standard deviation. Therefore, it is important to note that the observed differences between the two architectures may be of no statistical significance.

### 3. Hypothesis Evaluation

The initial hypothesis was a 3D-GAN implementation utilizing ConvNeXt architectural modifications to its generator would largely outperform the original DCGAN model, resulting in a higher quality of 3D generations containing fewer artifacts and improved quantitative assessment rates. The table below depicts the experimental results challenging this hypothesis.

| Metric | DCGAN (Baseline) | ConvNeXt-DCGAN | Difference | Hypothesis Supported |
|---|---|---|---|---|
| FID (↓) | **419.49** | 423.73 | +4.24 (+1.01%) | ✗ No |
| IS Mean (↑) | 1.319 | **1.336** | +0.017 (+1.29%) | ✓ Partial |
| IS Std | 0.038 | 0.043 | +0.005 | — |

Table 2: Summary of Experimental Results.

The experimental results do not support the original hypothesis, which was the change to a ConvNeXt style generator would yield improvements in 3D object generation quality. Contrary to expectations, the ConvNeXt variant produced samples with a higher FID score revealing a poorer distributional alignment with the real data compared to the original. This result directly contradicts the hypothesis. While the ConvNeXt variant was able to improve on the original architecture's performance in Inception Score due to the small magnitude of improvement this improvement may not be statistically significant. All in all, the marginal differences observed as well as after a qualitative examination of random samples from both generators do not provide compelling evidence for reduced artifacting in the ConvNeXt-modified architecture. (Figures 1 and 2 depict these random samples.)


Figure 1: ConvNeXt DCGAN Samples & Figure 2: DCGAN Samples.

Based on the experimental findings, the initial hypothesis can be revised as follows: Direct adaptation of ConvNeXt architectural principles does not yield significant improvements in 3D-GAN performance for voxel-based object synthesis. Effective transfer of modern SOTA CNN design principles to 3D generative modeling may require more complex, comprehensive modifications, domain-specific adaptations or alternative normalization strategies to account for the unique characteristics of adversarial training.

## 4. Conclusion

The experimental results indicate that replacing BatchNorm with LayerNorm in a ConvNeXt-style architecture does not yield substantial improvements in generative performance for 3D voxel synthesis, as measured by FID and IS metrics. Several factors may contribute to these findings, namely domain mismatch, normalization dynamics, and limited sample size. The ConvNeXt design principles, originally developed for discriminative tasks on natural images, may not directly transfer to generative modeling of 3D volumetric data. In the case of normalization dynamics, BatchNorm's batch-dependent statistics may provide implicit regularization benefits during adversarial training that are not replicated by LayerNorm. Finally, the topic of limited sample size: The evaluation was conducted on 500 generated samples, which may introduce variance in the computed metrics.

Comparative analysis reveals that both architectures achieve similar performance on the evaluated metrics, with the original DCGAN demonstrating a slight advantage in FID while the ConvNeXt variant shows marginal improvement in Inception Score. These results suggest that further architectural modifications or hyperparameter optimization may be necessary to achieve meaningful performance gains in 3D generative modeling tasks.

## 5. References


1. xchhuang, “simple-pytorch-3dgan/volumetric_data/chair/30/train at master · xchhuang/simple-pytorch-3dgan,” GitHub, 2025. https://github.com/xchhuang/simple-pytorch-3dgan/tree/master/volumetric_data/chair/30/train (accessed Dec. 09, 2025).
2. Atakan Erdoğan, “ConvNeXt — Next Generation of Convolutional Networks,” Medium, Apr. 2023. https://medium.com/@atakanerdogan305/convnext-next-generation-of-convolutional-networks-325607a08c46